

1. Basic Plotting

• Import: import matplotlib.pyplot as plt

• Simple Plot:

```
plt.plot(x, y, color='blue', linestyle='-', linewidth=1.5, marker='o', markersize=4, label='Series 1')
plt.show()
```

• Scatter Plot:

```
plt.scatter(x, y, s=20, c='red', alpha=0.7, label='Data Points')
```

- S: marker size
- c: color
- alpha: transparency

• Bar Plot:

```
plt.bar(categories, values, width=0.8, color='skyblue', edgecolor='black', align='center')
```

- width: width of bars
- edgecolor: border color of bars
- align: 'center' or 'edge'

• Histogram:

```
plt.hist(data, bins=10, color='green', alpha=0.8, density=True, edgecolor='black', histtype='bar')
```

- bins: number of bins or sequence of bin edges
- density: normalize to form a probability density
- histtype: 'bar', 'barstacked', 'step', 'stepfilled'

• Pie Chart:

```
plt.pie(sizes, labels=labels, autopct='%1.1f%%', colors=['gold', 'lightcoral', 'lightskyblue'], explode=(0, 0.1, 0), shadow=True, startangle=90)
```

- autopct: format string for percentage values
- explode: tuple of fractions for wedges to explode
- shadow: boolean, draw a shadow beneath the pie
- startangle: angle at which the first wedge is drawn

• Line Styles: '-' (solid), '--' (dashed), ':' (dotted), '-.-' (dash-dot)

• Markers: 'o' (circle), 'x' (x), '^' (triangle up), 's' (square), '+' (plus), 'D' (diamond)

• Colors: 'b' (blue), 'g' (green), 'r' (red), 'c' (cyan), 'm' (magenta), 'y' (yellow), 'k' (black), 'w' (white) or hex codes '#FF5733'

2. Figure and Axes

• Create Figure & Axes:

```
fig, ax = plt.subplots(figsize=(8, 6), dpi=100)
```

- figsize: tuple (width, height) in inches
- dpi: dots per inch for the figure

• Multiple Subplots:

```
fig, axes = plt.subplots(nrows=2, ncols=2, sharex=True, sharey=False)
```

- sharex: boolean, share x-axis between subplots
- sharey: boolean, share y-axis between subplots

• Adjust Layout:

```
plt.tight_layout(pad=1.0, w_pad=0.5, h_pad=1.0)
```

- pad: padding between the figure edge and the subplots
- w_pad: padding between subplots in width direction
- h_pad: padding between subplots in height direction

3. Customizing Plots

• Titles & Labels:

```
plt.title('Plot Title', fontsize=16, color='navy', loc='center')
plt.xlabel('X-axis Label', fontsize=12, color='darkgreen', labelpad=10)
plt.ylabel('Y-axis Label', fontsize=12, color='darkred', labelpad=10)
```

(or ax.set_title(), ax.set_xlabel(), ax.set_ylabel())

- fontsize: font size of the text
- color: color of the text
- loc: 'left', 'center', 'right' for title
- labelpad: spacing between label and axis

• Legend:

```
plt.legend(loc='upper right', ncol=1, fancybox=True, shadow=True, borderpad=1, labelspacing=0.5, fontsize='medium', frameon=True, framealpha=0.8, facecolor='lightgray', edgecolor='black')
```

(or ax.legend())

- loc: location string (e.g., 'best', 'upper left', 'lower center')
- ncol: number of columns in legend
- fancybox: boolean, round the corners of the legend box
- shadow: boolean, draw a shadow behind the legend box

- borderpad: padding between legend border and content
- labelspacing: vertical space between the legend entries
- fontsize: font size (e.g., 'small', 'large', or numeric)
- frameon: boolean, draw a frame around the legend
- framealpha: transparency of the legend frame
- facecolor: background color of the legend frame
- edgecolor: border color of the legend frame

• Limits:

```
plt.xlim(xmin, xmax)
plt.ylim(ymin, ymax)
```

(or ax.set_xlim(), ax.set_ylim())

• Grid:

```
plt.grid(True, linestyle='--', alpha=0.6, color='gray', linewidth=0.7)
```

(or ax.grid())

- linestyle: style of the grid lines
- alpha: transparency of the grid lines
- color: color of the grid lines
- linewidth: width of the grid lines

• Text Annotation:

```
plt.text(x_coord, y_coord, 'Annotation', fontsize=10, color='purple', ha='center', va='bottom', rotation=0, bbox=dict(facecolor='yellow', alpha=0.5))
```

- ha: horizontal alignment ('left', 'center', 'right')
- va: vertical alignment ('top', 'center', 'bottom', 'baseline')
- rotation: rotation angle of the text
- bbox: dictionary for text box properties

4. Advanced Plotting

• Logarithmic Scale:

```
plt.xscale('log', base=10)
plt.yscale('log', base=2)
```

(or ax.set_xscale(), ax.set_yscale())

- base: base of the logarithm

• Error Bars:

```
plt.errorbar(x, y, yerr=y_error, xerr=x_error, fmt='o', ecolor='red', capsize=3, capthick=1.5, elinewidth=1)
```

- yerr, xerr: magnitude of errors
- fmt: format string for data points
- ecolor: color of error bars
- capsize: length of error bar caps
- capthick: thickness of error bar caps
- elinewidth: thickness of error bar lines

5. Saving Plots

• Save Figure:

```
plt.savefig('my_plot.png', dpi=300, bbox_inches='tight', pad_inches=0.1)
```

- dpi: dots per inch
- bbox_inches: 'tight' to remove extra whitespace
- pad_inches: padding around the figure when bbox_inches='tight'

6. Configuration & Styles

• Runtime Configuration (rc_params):

```
plt.rcParams.update({
    'figure.figsize': [10, 6],
    'axes.labelsize': 14,
    'axes.titlesize': 16,
    'xtick.labelsize': 10,
    'ytick.labelsize': 10,
    'legend.fontsize': 10,
    'lines.linewidth': 2,
    'lines.markersize': 6,
    'font.family': 'serif',
    'font.serif': ['Times New Roman', 'Palatino', 'DejaVu Serif'],
    'savefig.facecolor': 'white'
})
```

7. Common Plot Types (Parameters)

Plot Type	Function	Key Parameters
Line Plot	plt.plot()	color, linestyle, linewidth, marker, markersize, label, alpha
Scatter Plot	plt.scatter()	S (size), C (color), marker, alpha, edgecolors, linewidths, label
Bar Chart	plt.bar()	width, color, edgecolor, align, alpha, bottom (for stacked bars)

Plot Type	Function	Key Parameters
Histogram	plt.hist()	bins, color, alpha, density, edgecolor, histtype, orientation
Box Plot	plt.boxplot()	vert, patch_artist, showmeans, meanline, flierprops, boxprops, medianprops
Pie Chart	plt.pie()	labels, autopct, colors, explode, shadow, startangle, wedgeprops
Image Display	plt.imshow()	cmmap (colormap), vmin, vmax (color limits), aspect, origin ('upper' or 'lower'), interpolation

8. Ticks and Spines

• Set Ticks:

```
ax.set_xticks([0, 1, 2])
ax.set_yticks([0, 0.5, 1])
```

• Set Tick Labels:

```
ax.set_xticklabels(['A', 'B', 'C'], rotation=45, ha='right', fontsize=9)
ax.set_yticklabels(['Low', 'Mid', 'High'], color='darkblue')
```

- rotation: angle of labels
- ha: horizontal alignment

• Tick Parameters:

```
ax.tick_params(axis='x', direction='inout', length=6, width=2, colors='r', grid_color='r', grid_alpha=0.5, top=True, labeltop=False, bottom=True, labelbottom=True)
```

- axis: 'x', 'y', or 'both'
- direction: 'in', 'out', or 'inout'
- length: length of the tick lines
- width: width of the tick lines
- colors: color of the tick lines and labels
- labeltop, labelbottom, labelleft, labelright: show/hide labels

• Spines:

```
ax.spines['right'].set_visible(False)
ax.spines['left'].set_linewidth(2)
ax.spines['bottom'].set_color('blue')
ax.spines['top'].set_linestyle('--')
```